Ebrahim Sharifi, Ph.D., P.Eng.

Ph.D. Industrial Engineer | Business and Supply Chain Analytics

Toronto, ON, Canada • +1 (647) 212-9446 • esharifi1372@gmail.com

LinkedIn • Personal Website • Google Scholar

SUMMARY

Ph.D. industrial engineer with expertise in business analytics, predictive modeling, and supply chain optimization. Taught **800+ students** across data mining, predictive modeling, and quantitative methods at Cape Breton University, earning **4.5 to 4.8 / 5** ratings. Industry background in supply chain analysis and project management. Seeking roles in business analytics, data analytics, or supply chain analytics in Canada.

CORE SKILLS

Machine Learning	XGBoost, Random Forest, Decision Trees, Naive Bayes, KNN, Neural Networks (ANN), K-Means Clustering, Classification, Regression, Feature Engineering, Model Evaluation
Tools and Languages	Python (pandas, scikit-learn), R, SQL, Excel, Tableau, MATLAB, GAMS, CPLEX
Analytics Methods	Predictive Modeling, Data Mining, Statistical Analysis, Forecasting, Optimization, Simulation, Multi-Criteria Decision Analysis, Time-Series Analysis
Supply Chain and Operations	Demand Forecasting, Inventory Management, Supplier Evaluation, Network Design, Robust Optimization, Stochastic Programming, Fuzzy Logic
Business Domains	Supply Chain and Logistics, Healthcare Analytics, E-commerce and Retail, Telecommunications, Manufacturing, Project Management
Communication	Stakeholder Reporting, Executive Briefings, Teaching and Mentorship, Cross-Functional Collaboration, Public Speaking

PROFESSIONAL EXPERIENCE

Assistant Professor (Business Analytics), Cape Breton University, Shannon School of Business — *Sep 2022 to Apr 2026*

Sydney, NS, Canada

- Designed and delivered four post-baccalaureate analytics courses (Data Mining, Data Visualization, Predictive Modeling, Quantitative Methods) across **6 sections/year** with 40–60 students each, plus 3 additional sections. Cumulative: **800+ students** over three years.
- Earned consistent **4.5 to 4.8 out of 5** on official student evaluations (Fall 2025, n=45, 85% response rate). Top dimension: returning assignments in a timely manner (**4.77 / 5**).
- Supervised **100+ final group projects** on real datasets (Kaggle, etc.) in healthcare, e-commerce, telecom. Coached teams through full ML pipelines: data cleaning, feature engineering, model training, evaluation, presentations using **XGBoost, Random Forest, Decision Trees, Naive Bayes, KNN, ANN, K-Means** in Python and R.
- Co-supervised graduate research; co-authored applied analytics work with **460+ citations** on Google Scholar (h-index 7).

Research Assistant (Operations and Decision Analytics), Toronto Metropolitan University (formerly Ryerson) — *Sep 2020 to Nov 2023*

Toronto, ON, Canada

- Built **multi-objective optimization and predictive models** for sustainable supply chain decisions on wheat, soybean, and biomass case studies. Presented at **Canadian Operations Research Society (CORS), INFORMS, IISE** conferences and chaired Supply Chain Optimization session. Published in top-tier journals (IF up to 12.1).

Project Manager, Mehr Naghsh Afarinan Shayan Company — *Sep 2017 to Jul 2020*

Tehran, Iran

- Led a team of 5 to develop a **tourism mobile application** end-to-end: scope, scheduling, resource allocation, and delivery.
- Managed **stakeholder engagement and fund acquisition**, presenting business cases to investors and coordinating across technical and business teams to keep the project on track.

Supply Chain Analyst, Sazeh Gostar Saipa (S.G.S.), SAIPA Automotive Group *Sep 2016 to Sep 2017*

Tehran, Iran

- Applied **multi-criteria decision analysis (MCDA)** to evaluate replacement vehicle for import/assembly (technical, cost, market criteria) and conduct market analysis on bus procurement from China. Automated recurring reporting processes across strategic planning centre.

SELECTED PROJECTS

Sustainable Wheat Supply Chain Network Design Under Uncertainty

First to integrate sustainability, resiliency, and responsiveness simultaneously. Modeled real Ontario wheat network (4 farms, 4 elevators, 5 DCs, 10 demand zones) with wheat quality/blending, intermodal transport, winter weather (–30% rail capacity), and MRE policy constraints. Optimized across four objectives: cost, delivery time, environmental/social impact, and resiliency. Key result: intermodal transport reduced costs by 8% and delivery times by 20% compared to truck-only, while maintaining supply chain resilience under disruptions. Published in Journal of Cleaner Production (IF 9.8).

Methods and tools: Multi-objective optimization, Fuzzy Robust Stochastic programming, Best-Worst Method, IBM CPLEX

Sustainable Soybean Supply Chain Design Under Uncertainty (Canadian Case Study)

Two-stage optimization framework for Canadian soybean supply chain (Ontario/Quebec/Manitoba to China/Europe/USA). Ranked suppliers across 20 sustainability criteria, then optimized four competing objectives: profit, job creation, supplier sustainability, and CO_2 emissions under demand/price uncertainty with 18 scenarios. Modeled currency exchange rates and customs duties for international trade. Key result: integrating supplier sustainability into network design changed supplier selection and reduced profit by 12%, but government SDG compliance incentives offset this gap, making sustainable sourcing economically viable. Published in Sustainable Production and Consumption (Elsevier, Vol. 40, IF 12.1).

Methods and tools: Two-stage stochastic optimization, IT2TrF-BWM, augmented ε-constraint, IBM CPLEX

Advancing Minority Class Detection in Diabetic Diagnosis

ML pipeline for multiclass diabetes prediction (healthy/prediabetic/diabetic) using CDC BRFSS dataset (253,680 records, 21 features, 84.2% class imbalance). Evaluated 6 resampling strategies across 5 classifiers to address severe underrepresentation of diabetic and prediabetic classes. Applied Borda count consensus ranking for robust feature selection. Key result: Tomek Links paired with lightweight boosting (LightGBM, CatBoost) achieved best performance for early diabetes detection, with 15% improvement in recall for minority classes compared to baseline models. Emphasis on minimizing false negatives to support clinical decision-making.

Methods and tools: Optuna hyperparameter tuning, Borda count feature selection, SMOTE/Tomek Links resampling, LightGBM, CatBoost, Python

EDUCATION

Ph.D., Industrial Engineering *2020 to 2023*

Toronto Metropolitan University, Toronto, ON. Grade: A+ (GPA 4.33 / 4.33). Completed in 3 years. Nominated for the Governor General's Gold Medal.

M.Sc., Industrial Engineering (Engineering Management) *2016 to 2018*

Iran University of Science and Technology. GPA: 90 / 100.

CERTIFICATIONS

- **Data Science Fundamentals with Python and SQL** (5-course specialization), IBM, 2026
- **Advanced Business Analytics** (5-course specialization), University of Colorado Boulder, 2026
- **Professional Engineer (P.Eng.)**, Professional Engineers Ontario (PEO), 2026
- **Diploma in Operational research**, Canadian Operations Research Society (CORS), 2021

HONOURS AND AFFILIATIONS

- Nominated for **Governor General's Gold Medal** for outstanding dissertation (highest academic honour in Canada).
- TMU Doctoral Tuition Scholarship (3 years full tuition + $24,000)
- Toronto Met Graduate Scholarship ($15,000),
- MIE Graduate Studies Award ($2,000).
- **Professional Membership:** CORS, INFORMS, ASAC, PEO.